

Truly Innovative **TECH**

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Tech*

In-cabin laser flood illuminators

The new TARA2000-AUT family of vertical-cavity surface-emitting laser (VCSEL) flood illuminators are ideal for new optical in-cabin sensing (ICS) systems based on 2D NIR imaging or 3D time-of-flight sensing, which support the next generation of assisted and autonomous driving technologies



in vehicles. Driver monitoring systems track whether the driver is alert and attentive while driving. A high optical power is distributed over the whole field of illumination (FoI). Now a single TARA2000-AUT illuminator can replace multiple lower-power emitters, offering automotive OEMs savings in space, component count, and cost along with high immunity to interference by sunlight and improved system performance. It is also available with an ultra-wide FoI to give excellent responsiveness in gesture-sensing applications for use by both the driver and passengers.

ams
www.ams.com

Small, rugged photocouplers

Measuring a mere 2.5mm x 2.1mm in an LSS05 package, the RV1S9231A 2.5 A output and RV1S9207A 0.6 A output IGBT drivers, and RV1S9209A active high-output IPM drivers come in low-pro-



file LSS05 packages with a 0.65mm pin pitch, 5,000Vrms isolation, and high temperature operation up to 125°C. The small optical isolated IGBT drivers and intelligent power module drivers reduce PCB mounting areas by up to 35% compared to other devices. The photocoupler trio also supports 200V and 400V systems with reinforced insulation to meet stringent industrial safety standards, adhering to the strict UL61800-5-1 standards for motor drive equipment.

Renesas Electronics Corporation
www.renesas.com

AI Edge intelligent devices

The Versal AI Edge series enables AI innovation from the edge to the endpoint. With 4x the AI performance-per-watt versus GPUs and 10x greater compute density versus previous-generation adap-



tive SoCs, the Versal AI Edge series claims to be the world's most scalable and adaptable portfolio for next-generation distributed intelligent systems. Versal AI Edge adaptive compute acceleration platforms deliver intelligence to a wide range of applications including automated driving with the highest levels of functional safety, collaborative robotics, predictive factory and healthcare systems, and multi-mission payloads for the aerospace and defence markets. Versal AI Edge devices achieve high compute density by delivering 10x compute density for more intelligent autonomous systems. Additionally, Versal AI Edge devices support multiple safety standards across industrial, avionics, and automotive markets.

Xilinx
www.xilinx.com

Graphene Hall Effect sensor

The graphene GHS01AT Hall Effect sensor is optimised for use in relatively low field environments and normal ambient temperatures. Bringing the magnetic field measurement resolution towards that of more complex magnetic



sensors, yet with the small size and ease of use of a Hall sensor, it can address monitoring tasks for which conventional technologies are not able to provide an effective solution. The sensors are ideally suited for battery cell analysis for validity of different battery cell chemistry derivatives and form factors under development and thus enable detailed and point-to-point understanding of battery cell behaviour. If hotspots arise, the local mapping of internal cell resistance in these areas could provide insights into the physical processes occurring in the lead up to their formation.

Paragraf
www.paragraf.com

Ultra-long-range lidar

Aeva Technologies' 4D lidar now delivers a breakthrough development that enables instant detection and identification



of pedestrians beyond 350m and dark objects, including vehicles beyond 500m. This is a step-change in capability in long-range sensing and has the potential to provide crucial safety enhancements for